BASIC ELECTRONICS ENGINEERING • PART B

UNIT II

Basic Electronic Circuits & Instrumentation

Rectifiers, filters, regulators, public address system, RC-coupled amplifier and instrumentation

WHAT YOU WILL LEARN

1 Block diagram of a DC power supply
2 Half-wave & full-wave bridge rectifier
3 Capacitor filter
4 Zener voltage regulator
5 Public Address (PA) system
6 RC-coupled CE amplifier & response
7 Electronic instrumentation system
8 Summary & exam questions

1 DC Power Supply: Block Diagram

Most electronic circuits need **steady DC**, but the wall socket gives **AC**. A DC power supply converts one into the other in five simple stages.

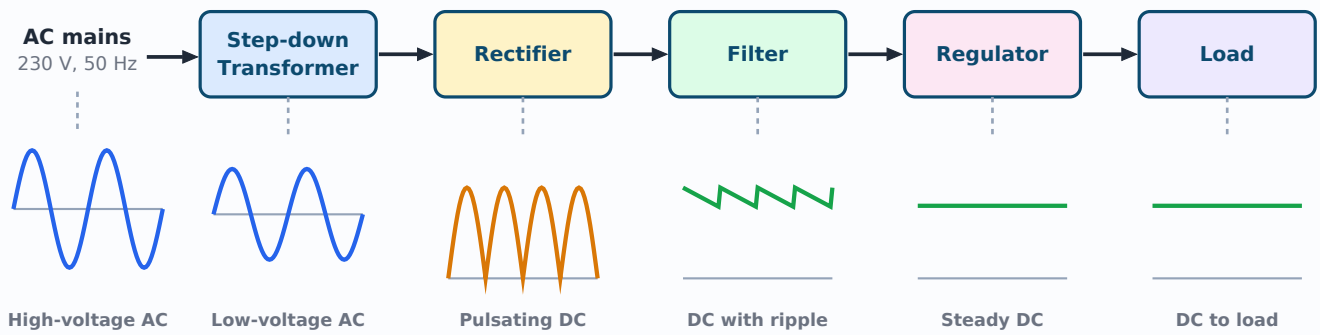


Fig 2.1 Block diagram of a regulated DC power supply, with the waveform at each stage

Block	Function
Step-down transformer	Reduces the 230 V AC mains to a safe, low voltage (e.g. 12 V) and gives electrical isolation
Rectifier	Converts AC to **pulsating DC** using diodes (they conduct in one direction only)
Filter	Smooths the pulsations (removes most of the ripple), usually using a capacitor
Regulator	Keeps the output voltage **constant** despite changes in mains voltage or load (e.g. Zener / IC 7805)
Load	The electronic circuit or device being powered

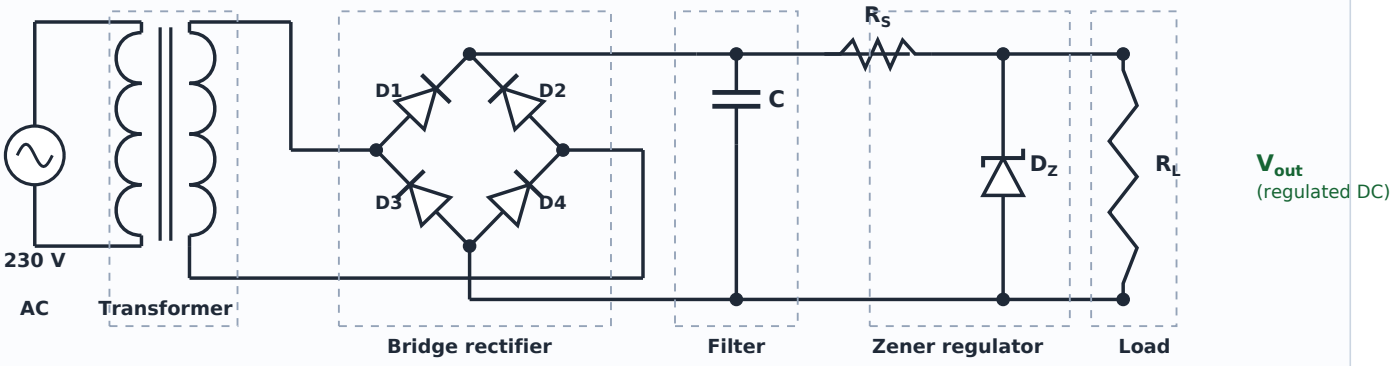


Fig 2.2 Complete regulated power supply circuit (transformer, bridge rectifier, capacitor filter and Zener regulator)

2 Rectifiers: Half-Wave and Full-Wave Bridge

Starting point: the half-wave rectifier

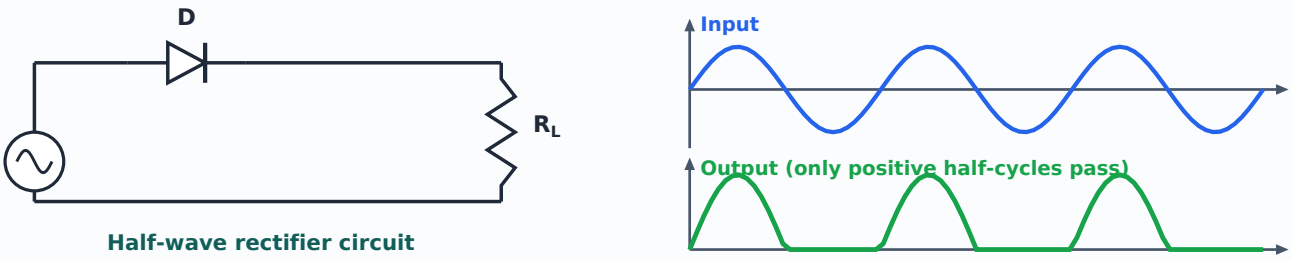


Fig 2.3 Half-wave rectifier circuit and waveforms

- One diode conducts only in the **positive half-cycle**; during the negative half-cycle it is reverse biased and the output is zero.
- $\mathbf{V_{dc} = V_m/\pi \approx 0.318\ V_m}$; ripple factor 1.21; efficiency 40.6 %; PIV = V_m; ripple frequency = f.
- Drawback: half of the input is wasted and the output is very rough, so we use **full-wave** rectification.

Full-wave bridge rectifier

DEFINITION
A **rectifier** converts AC into unidirectional (pulsating DC) current. The **bridge rectifier** uses **four diodes** in a diamond so that **both half-cycles** of the AC input reach the load in the **same direction**, and it needs no centre-tapped transformer.

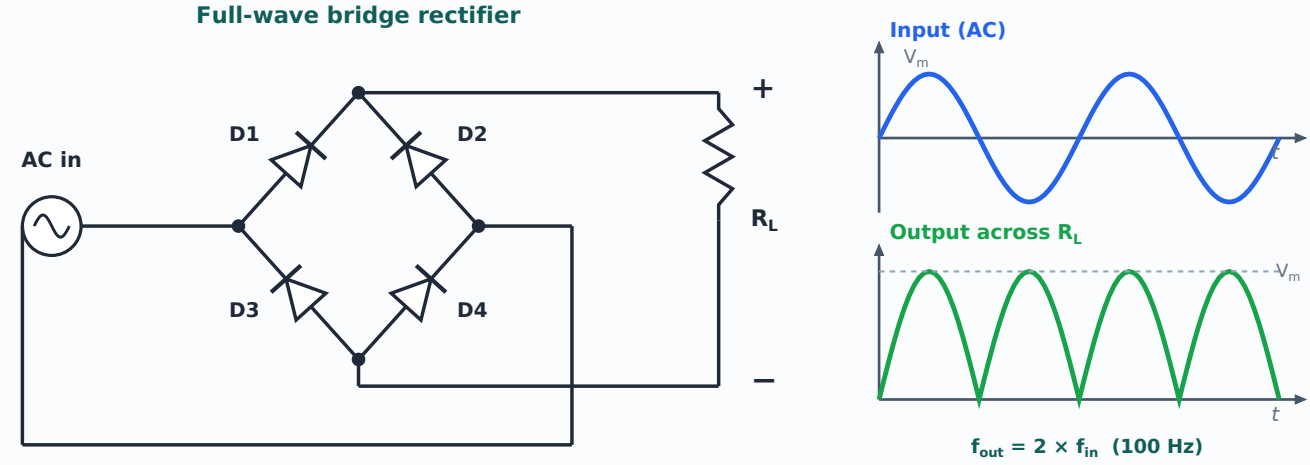


Fig 2.4 Bridge rectifier circuit and its input/output waveforms

Working

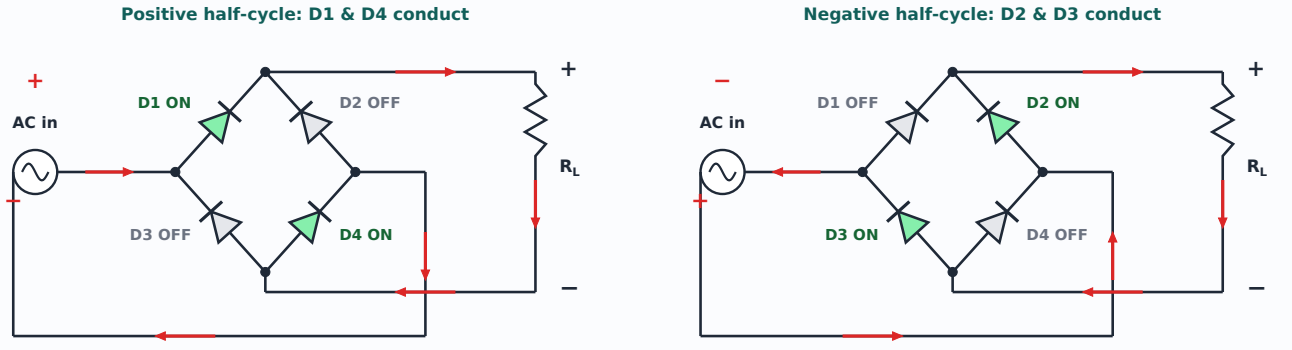


Fig 2.5 Current path in the two half-cycles (red arrows: direction of current; green diodes: conducting)

Input half-cycle	Diodes ON	Diodes OFF	Current through R_L
Positive (top end +)	D_1 and D_4	D_2 and D_3	Top to bottom
Negative (top end −)	D_2 and D_3	D_1 and D_4	**Top to bottom (same direction)**

Important formulas (ideal diodes, peak secondary voltage V_m)
$V_{dc} = 2V_m/\pi \approx 0.637\ V_m$ | $V_{rms} = V_m/\sqrt{2}$ | $I_{dc} = 2I_m/\pi$
Ripple factor $\gamma \approx 0.48$ | Efficiency $\eta = 81.2\ \%$ | PIV $= V_m$ | Ripple frequency $= 2f$

Parameter	Half-wave	Centre-tap full-wave	Bridge full-wave
Number of diodes	1	2	4
V_{dc}	V_m/π	$2V_m/\pi$	$2V_m/\pi$
PIV of each diode	V_m	$2V_m$	**V_m**
Ripple factor	1.21	0.48	0.48
Efficiency	40.6 %	81.2 %	81.2 %
Ripple frequency	f	2f	2f
Centre-tap transformer	No	**Yes**	**No**

Performance term	Meaning
Ripple factor (γ)	Ratio of AC (ripple) component to DC component in the output; smaller is better ($\gamma = V_{r(rms)} / V_{dc}$)
PIV (peak inverse voltage)	Maximum reverse voltage that a diode must withstand without breaking down
Efficiency (η)	DC output power / AC input power × 100 %
Regulation	How little the output voltage changes with load current (and input voltage)

Example
Secondary = 12 V rms. $V_m = 12 \times 1.414 = 16.97$ V. Ideal $V_{dc} = 0.637 \times 16.97 \approx$ **10.8 V**.
In practice two diodes (2 × 0.7 V = 1.4 V) conduct at a time: $V_{dc} \approx 0.637 \times (16.97 - 1.4) \approx$ **9.9 V**. Each diode needs PIV ≥ 17 V (a 1N4007 is ample).

Why the bridge is preferred
- Uses the whole transformer winding all the time (better utilisation), PIV across each diode is only V_m.
- Output ripple is at 2f (100 Hz), which is easier to filter than f (50 Hz) in a half-wave rectifier.
- Disadvantage: two diode drops (≈1.4 V) in the path, so a little more power loss at low voltages.

3 Capacitor Filter

The rectifier output is DC in direction but not steady: it pulses at 100 Hz. A **capacitor filter** (a large electrolytic capacitor connected **in parallel** with the load) stores charge at the peaks and releases it in the gaps.

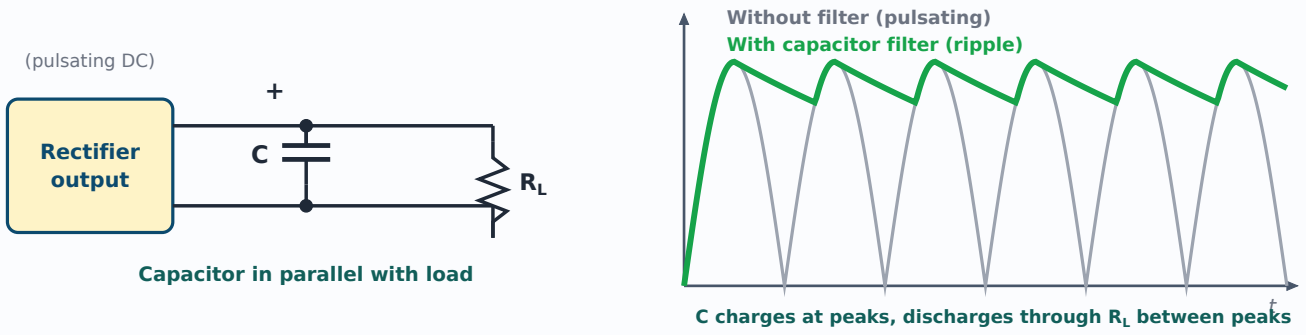


Fig 2.6 Capacitor filter and the smoothed output voltage

- **Rising part:** when the rectified voltage rises above the capacitor voltage, the diodes conduct and the capacitor **charges** quickly to the peak V_m.
- **Falling part:** the input drops below the capacitor voltage, the diodes switch OFF and the capacitor **discharges slowly** through R_L (time constant $R_L C$) so the load voltage barely falls.
- **Next peak:** the capacitor is topped up again, which gives a nearly flat DC with a small **ripple**.

Effects of the capacitor filter

- Larger C or larger R_L (lighter load) → **less ripple** and higher V_{dc} (closer to V_m).
- Heavy load (small R_L) → capacitor drains faster → **more ripple**.
- Typical values: 470 µF to 4700 µF electrolytic; observe correct polarity and a voltage rating above V_m.

4 Simple Zener Voltage Regulator

Even after filtering, the DC output changes when the mains voltage or the load changes. A **regulator** keeps V_{out} constant. The simplest one uses a **Zener diode in reverse breakdown** in parallel with the load, plus a series resistor R_S.

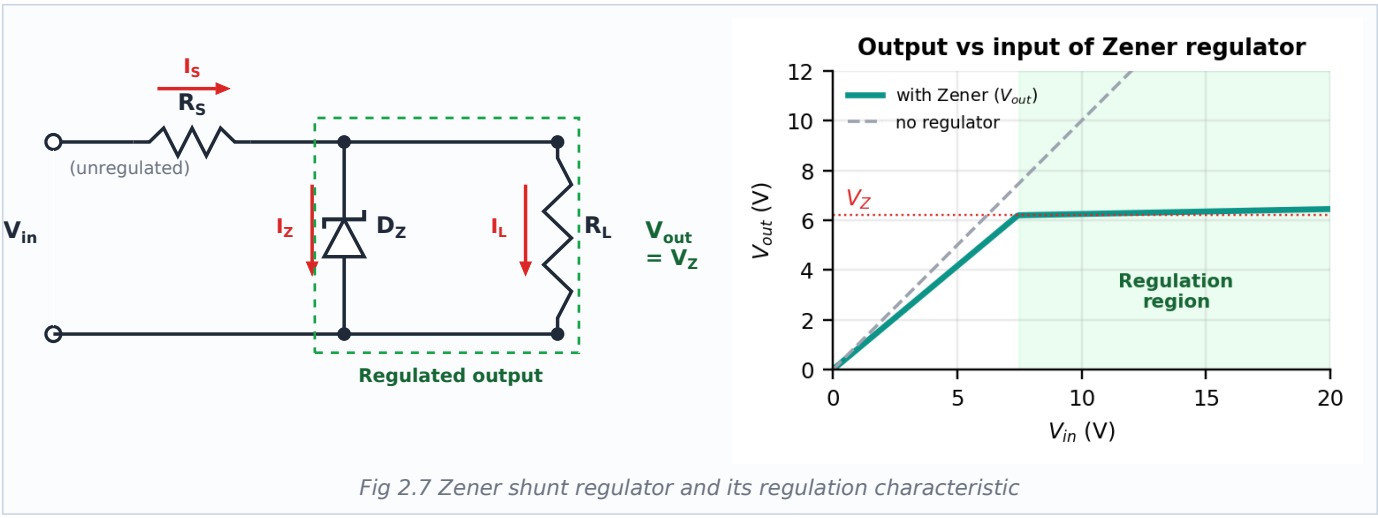


Fig 2.7 Zener shunt regulator and its regulation characteristic

Working

- The Zener diode is **reverse biased** and operates in breakdown, so the voltage across it (and the load) stays at $\mathbf{V_Z}$.
- **If V_{in} increases:** I_S rises; the extra current flows through the Zener (I_Z ↑). The extra drop appears across R_S, so V_{out} stays at V_Z.
- **If load current I_L increases:** I_Z decreases by the same amount, I_S stays almost constant, so V_{out} stays at V_Z.
- **Rule:** I_Z must stay between $I_{Z(min)}$ and $I_{Z(max)}$ for regulation to hold.

Design equations

$\mathbf{I_S = I_Z + I_L = (V_{in} - V_Z) / R_S}$ | $\mathbf{V_{out} = V_Z}$ | $\mathbf{P_{Z(max)} = V_Z\, I_{Z(max)}}$

Line regulation = $\Delta V_{out}/\Delta V_{in} \times 100\ \%$ | Load regulation = $(V_{NL} - V_{FL})/V_{FL} \times 100\ \%$

Example

V_{in} = 12 V, V_Z = 5.1 V, R_L = 1 kΩ → I_L = 5.1 mA. For I_Z = 5 mA: $R_S = (12 - 5.1)/(5 + 5.1)$ mA = **683 Ω ≈ 680 Ω**.

Limitations

Wastes power in R_S and the Zener, fixed output, suitable only for small load currents. Higher currents use transistor or IC regulators (e.g. 7805).

5 Amplifiers and the Public Address (PA) System

DEFINITION

An **amplifier** increases the strength (voltage, current or power) of a signal without changing its shape. Voltage gain $A_v = V_o/V_i$; in decibels, **gain (dB) = 20 $\log_{10} A_v$**.

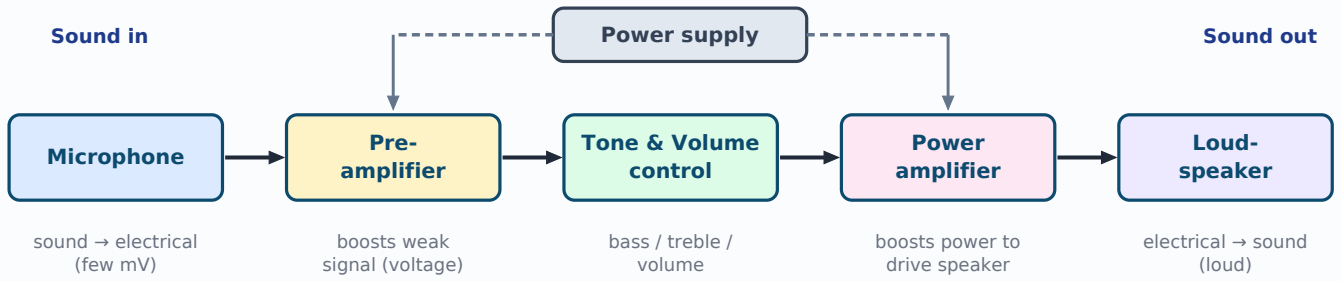


Fig 2.8 Block diagram of a Public Address (PA) system

Block	Role
Microphone	Transducer: converts sound waves into a tiny electrical signal (a few mV)
Pre-amplifier	Low-noise voltage amplifier that raises the weak signal to a usable level
Tone / volume control	Adjusts bass, treble and loudness; shapes the signal before power amplification
Power amplifier	Supplies high current/power needed to drive the loudspeaker
Loudspeaker	Transducer: converts the electrical signal back to sound
Power supply	Gives DC to the amplifier stages

Voltage gain A_v	1	2	10	100	1000	0.707
Gain in dB	0	6 dB	20 dB	40 dB	60 dB	−3 dB

Acoustic feedback (“howling”)

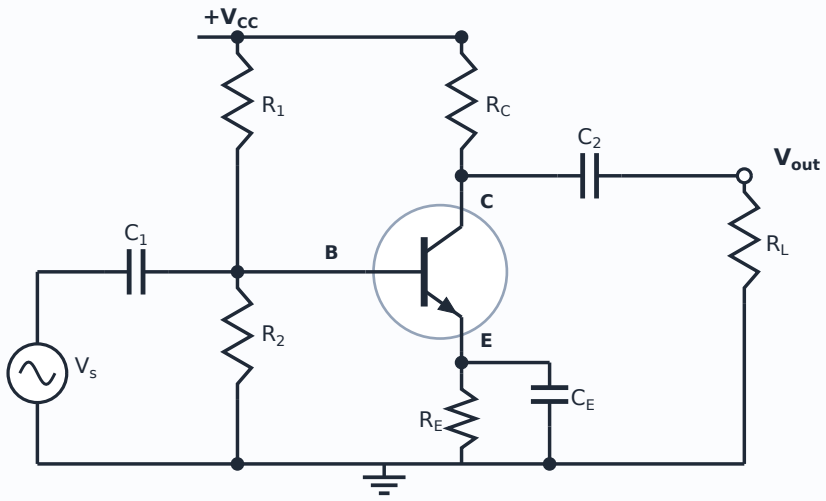
If the loudspeaker sound re-enters the microphone, it is amplified again and again, producing a loud whistle. Cure: keep the microphone away from the speakers, lower the gain or use directional microphones.

APPLICATIONS

Railway stations, airports, auditoriums, classrooms, stadiums, religious places, emergency announcements.

6 RC-Coupled CE Amplifier and its Frequency Response

When one transistor stage cannot give enough gain, stages are cascaded. They are linked by a **coupling network**; the most common is **RC coupling**, where a capacitor passes the AC signal and blocks the DC of the previous stage.



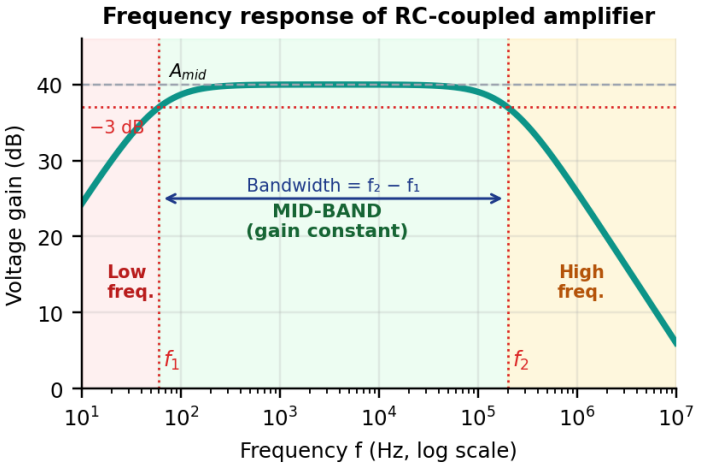
Components

- **R_1, R_2** – voltage-divider bias
- **R_C** – collector (load) resistor
- **R_E** – bias stabilisation
- **C_E** – bypass: prevents gain loss
- **C_1, C_2** – coupling capacitors: pass AC, block DC
- **R_L** – next stage / load

Fig 2.9 Single stage of an RC-coupled common-emitter amplifier

Working

- **DC:** R_1, R_2, R_E fix the Q-point in the active region, so the transistor always conducts.
- **AC:** the input signal passes through C_1 to the base and varies I_B. The collector current varies β times more, producing a large signal voltage across R_C.
- The amplified signal, **180° out of phase** with the input, goes through C_2 to the load. C_2 blocks the DC at the collector.



Why does the gain change with frequency?

Low frequencies: reactance $X_C = 1/(2\pi fC)$ of C_1, C_2 and C_E is large, so part of the signal is dropped across them → gain falls.

Mid frequencies: capacitors behave like short circuits; transistor capacitances are negligible → **gain is maximum and constant**.

High frequencies: transistor junction and stray capacitances shunt the signal, and β falls → gain falls.

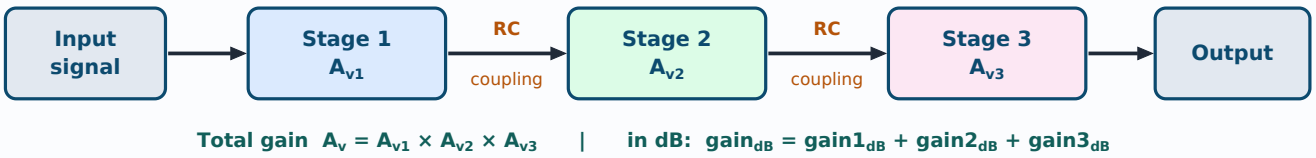
Fig 2.10 Frequency response curve (gain in dB against log frequency)

Bandwidth

The **half-power (−3 dB) frequencies** f_1 (lower) and f_2 (upper) are where the gain drops to **0.707 A_{mid}** ($A_{mid}/\sqrt{2}$). **Bandwidth BW = $f_2 - f_1$**. For audio amplifiers this is about 20 Hz to 20 kHz.

Advantages	Disadvantages
Simple, cheap and compact; no bulky transformers or inductors	Poor impedance matching, so low power transfer to a low-resistance load (e.g. a speaker)
Good, wide, flat response over the audio range	Gain falls at very low and very high frequencies
Little frequency distortion; easy to cascade many stages	Cannot amplify DC; power gain is modest (so used for voltage amplification)

Multistage amplifiers and coupling methods



Total gain $A_v = A_{v1} \times A_{v2} \times A_{v3}$ | **in dB:** $gain_{dB} = gain1_{dB} + gain2_{dB} + gain3_{dB}$

Fig 2.11 Cascaded amplifier stages joined by coupling networks

Coupling	How it works	Remark
RC coupling	Capacitor + resistor pass AC and block DC	Cheap, wide bandwidth: audio voltage amplifiers
Transformer coupling	Transformer transfers the signal and matches impedance	Good power transfer but bulky, costly; narrow band
Direct coupling	Output of one stage connected directly to the next	Can amplify DC; stage biasing interacts (drift)

7 Electronic Instrumentation System

DEFINITION

Electronic instrumentation is the use of electronic circuits to **measure, display, record and control** physical quantities (temperature, pressure, light, force, speed, heart-beat, ...) by first converting them into electrical signals.

Example: digital thermometer → heat ▶ thermistor ▶ amplifier ▶ ADC ▶ numeric display

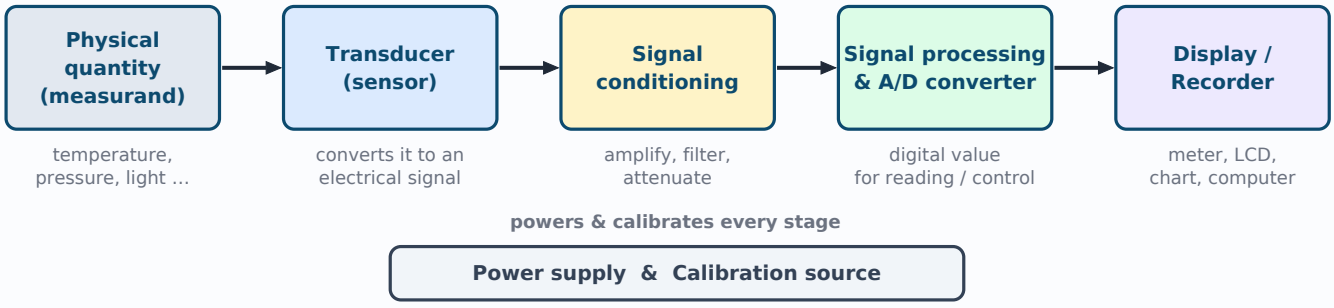


Fig 2.12 Block diagram of an electronic instrumentation system

Block	Function	Example
Transducer / sensor	Converts a physical quantity into an electrical signal	Thermocouple, thermistor, LDR, strain gauge, microphone
Signal conditioning	Amplifies, filters, attenuates or linearises the weak signal	Op-amp amplifier, low-pass filter
Signal processing / ADC	Converts analog to digital; computes, compares, controls	Microcontroller, ADC chip
Display / recorder	Presents the result for the user or stores/transmits it	LCD, LED, meter, chart recorder, PC
Power supply, calibration	Powers all blocks; reference to keep accuracy	Regulated DC supply

Common transducers

Quantity	Transducer	Principle
Temperature	Thermistor, thermocouple, RTD	Resistance / voltage changes with temperature
Light	LDR, photodiode	Resistance / current changes with light intensity
Pressure, force, weight	Strain gauge, load cell	Resistance changes when stretched
Displacement	LVDT, potentiometer	Induced voltage / resistance varies with position
Sound	Microphone	Sound pressure to voltage

Active vs passive transducers

- **Active** transducers generate their own voltage (thermocouple, piezo-electric crystal, photovoltaic cell).
- **Passive** transducers need an external supply (thermistor, strain gauge, LDR).

APPLICATIONS

Digital multimeter, oscilloscope, ECG / patient monitor, weighing machine, industrial process control, weather stations, smart-home sensors.

Advantages of electronic instruments

High sensitivity and speed, remote measurement, easy digital display and storage, small size, and the ability to connect to computers and control systems.

8 Quick Summary and Exam Corner

Unit II in one glance

- DC supply = **Transformer → Rectifier → Filter → Regulator → Load**.
- Bridge rectifier: 4 diodes; two conduct in each half-cycle; $V_{dc} = 2V_m/\pi$; PIV = V_m; ripple frequency = 2f; $\eta = 81.2$ %.
- Capacitor filter: C in parallel with load; charges at peaks, discharges slowly; larger C → smaller ripple.
- Zener regulator: reverse-biased Zener in parallel with load plus R_S; $V_{out} = V_Z$; $R_S = (V_{in} - V_Z)/(I_Z + I_L)$.
- PA system: Microphone → Pre-amp → Tone control → Power amp → Loudspeaker.
- RC-coupled CE amplifier: flat mid-band; gain falls at low f (coupling/bypass caps) and high f (junction/stray caps); BW = $f_2 - f_1$.
- Instrumentation: Measurand → Transducer → Signal conditioning → Processing/ADC → Display.

Short answer questions (2 marks)

1. List the blocks of a regulated DC power supply.
2. Why is a bridge rectifier preferred over a centre-tap rectifier?
3. What is the ripple frequency of a full-wave rectifier with 50 Hz input?
4. State the function of a filter and of a regulator.
5. What is the purpose of the bypass capacitor in a CE amplifier?
6. Define bandwidth of an amplifier.
7. What is a transducer? Give two examples.

Long answer questions (5 / 10 marks)

1. Draw the block diagram of a DC power supply and explain each block with waveforms.
2. With a neat circuit diagram, explain the working of a full-wave bridge rectifier. Give its V_{dc}, ripple factor, efficiency and PIV.
3. Explain the working of a capacitor filter with waveforms.
4. Explain the working of a Zener voltage regulator for variations in input voltage and load.
5. Draw the block diagram of a PA system and explain the function of each block.
6. Draw the circuit of an RC-coupled CE amplifier, explain its working and sketch its frequency response.

7. Draw and explain the block diagram of an electronic instrumentation system.

Numerical practice (with answers)

1. A bridge rectifier has a secondary of 20 V rms. Find V_m, ideal V_{dc} and PIV. *Ans: V_m = 28.3 V, V_{dc} = 18.0 V, PIV = 28.3 V*
2. A Zener (V_Z = 9.1 V) is fed from 15 V through R_S = 590 Ω. With no load, find the Zener current. *Ans: (15 − 9.1)/590 = 10 mA*
3. An amplifier has A_{mid} = 200, f_1 = 40 Hz, f_2 = 80 kHz. Find the bandwidth and the gain at the −3 dB points. *Ans: BW ≈ 79.96 kHz; gain = 200/√2 ≈ 141*
4. Convert a voltage gain of 100 into dB. *Ans: 20 log 100 = 40 dB*